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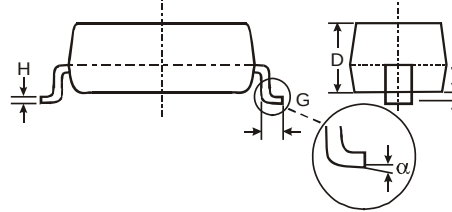
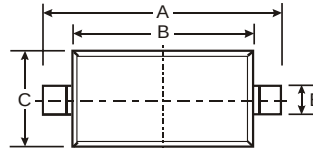
Jameco Part Number 1591156

Features

- 500mW Power Dissipation
- General Purpose, Medium Current
- Ideally Suited for Automated Assembly Processes
- **Lead Free/RoHS Compliant (Note 2)**
- **Qualified to AEC-Q101 Standards for High Reliability**

Mechanical Data

- Case: SOD-123
- Case Material: Molded Plastic. UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020D
- Terminals: Solderable per MIL-STD-202, Method 208
- Lead Free Plating (Matte Tin Finish annealed over Alloy 42 leadframe)
- Polarity: Cathode Band
- Ordering Information: See Page 4
- Marking Information: See Electrical Characteristics Table and Page 4
- Weight: 0.01 grams (approximate)



SOD-123			
Dim	Min	Max	Typ
A	3.55	3.85	3.65
B	2.55	2.85	2.65
C	1.40	1.70	1.55
D	1.00	1.35	1.05
E	—	—	0.55
G	0.25	0.40	0.30
H	0.10	0.15	0.11
J	—	0.10	0.05
α	0	8°	—
All Dimensions in mm			

Maximum Ratings @T_A = 25°C unless otherwise specified

Characteristic	Symbol	Value	Unit
Forward Voltage @ I _F = 10mA	V _F	0.9	V
Power Dissipation (Note 1)	P _D	500	mW
Thermal Resistance, Junction to Ambient Air (Note 1)	R _{θJA}	350	°C/W
Operating and Storage Temperature Range	T _J , T _{STG}	-65 to +150	°C

- Notes:
1. Device mounted on ceramic PCB; 7.6 mm x 9.4 mm x 0.87 mm with pad areas 25 mm².
 2. No purposefully added lead.

Electrical Characteristics

@T_A = 25°C unless otherwise specified

Type Number	Type Code	Zener Voltage Range (Note 4)			Test Current	Maximum Zener Impedance (Note 3)		Maximum Reverse Leakage Current (Note 4)	
		V _Z @ I _{ZT}			I _{ZT}	Z _{ZT} @ I _{ZT}	Z _{ZK} @ I _{ZK} = 0.25mA	I _R	@ V _R
		Nom (V)	Min (V)	Max (V)	mA	Ω		μA	V
MMSZ5221B	C1	2.4	2.28	2.52	20	30	1200	100	1.0
MMSZ5223B	C3	2.7	2.57	2.84	20	30	1300	75	1.0
MMSZ5225B	C5	3.0	2.85	3.15	20	30	1600	50	1.0
MMSZ5226B	G1	3.3	3.14	3.47	20	28	1600	25	1.0
MMSZ5227B	G2	3.6	3.42	3.78	20	24	1700	15	1.0
MMSZ5228B	G3	3.9	3.71	4.10	20	23	1900	10	1.0
MMSZ5229B	G4	4.3	4.09	4.52	20	22	2000	5.0	1.0
MMSZ5230B	G5	4.7	4.47	4.94	20	19	1900	5.0	2.0
MMSZ5231B	E1	5.1	4.85	5.36	20	17	1600	5.0	2.0
MMSZ5232B	E2	5.6	5.32	5.88	20	11	1600	5.0	3.0
MMSZ5233B	E3	6.0	5.70	6.30	20	7	1600	5.0	3.5
MMSZ5234B	E4	6.2	5.89	6.51	20	7	1000	5.0	4.0
MMSZ5235B	E5	6.8	6.46	7.14	20	5	750	3.0	5.0
MMSZ5236B	F1	7.5	7.13	7.88	20	6	500	3.0	6.0
MMSZ5237B	F2	8.2	7.79	8.61	20	8	500	3.0	6.5
MMSZ5238B	F3	8.7	8.27	9.14	20	8	600	3.0	6.5
MMSZ5239B	F4	9.1	8.65	9.56	20	10	600	3.0	7.0
MMSZ5240B	F5	10	9.50	10.50	20	17	600	3.0	8.0
MMSZ5241B	H1	11	10.45	11.55	20	22	600	2.0	8.4
MMSZ5242B	H2	12	11.40	12.60	20	30	600	1.0	9.1
MMSZ5243B	H3	13	12.35	13.65	9.5	13	600	0.5	9.9
MMSZ5245B	H5	15	14.25	15.75	8.5	16	600	0.1	11
MMSZ5246B	J1	16	15.20	16.80	7.8	17	600	0.1	12
MMSZ5248B	J3	18	17.10	18.90	7.0	21	600	0.1	14
MMSZ5250B	J5	20	19.00	21.00	6.2	25	600	0.1	15
MMSZ5251B	K1	22	20.90	23.10	5.6	29	600	0.1	17
MMSZ5252B	K2	24	22.80	25.20	5.2	33	600	0.1	18
MMSZ5254B	K4	27	25.65	28.35	5.0	41	600	0.1	21
MMSZ5255B	K5	28	26.60	29.40	4.5	44	600	0.1	21
MMSZ5256B	M1	30	28.50	31.50	4.2	49	600	0.1	23
MMSZ5257B	M2	33	31.35	34.65	3.8	58	700	0.1	25
MMSZ5258B	M3	36	34.20	37.80	3.4	70	700	0.1	27
MMSZ5259B	M4	39	37.05	40.95	3.2	80	800	0.1	30

Notes: 3. f = 1KHz.
4. Short duration pulse test used to minimize self-heating effect.

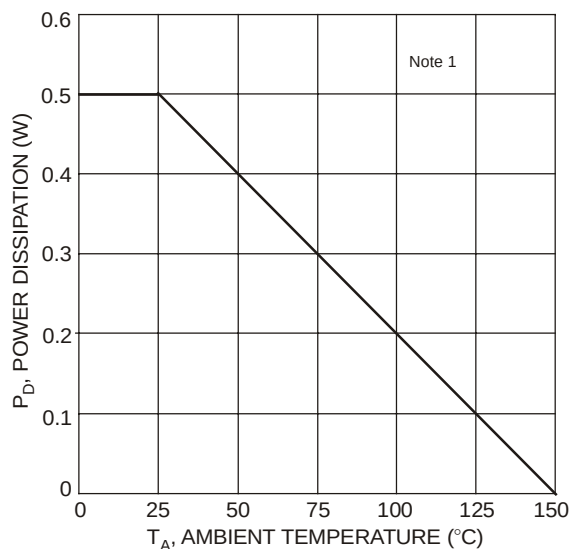


Fig. 1 Power Dissipation vs Ambient Temperature

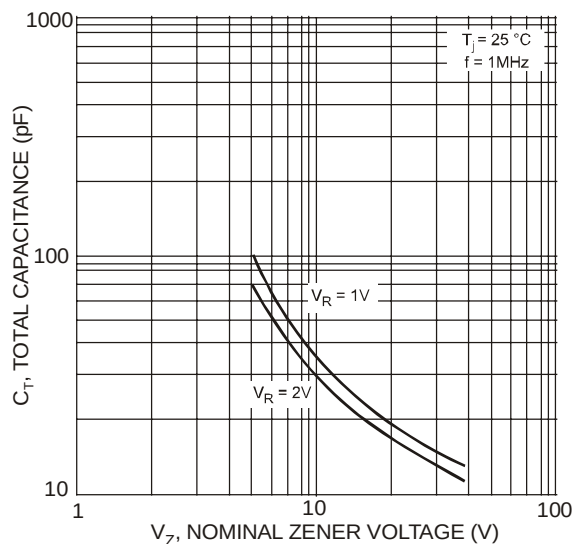


Fig. 2 Total Capacitance vs Nominal Zener Voltage

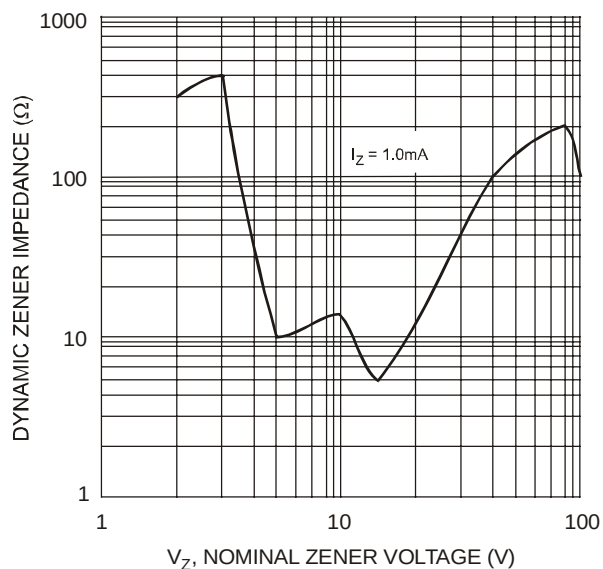


Fig. 3 Zener Voltage vs. Zener Impedance

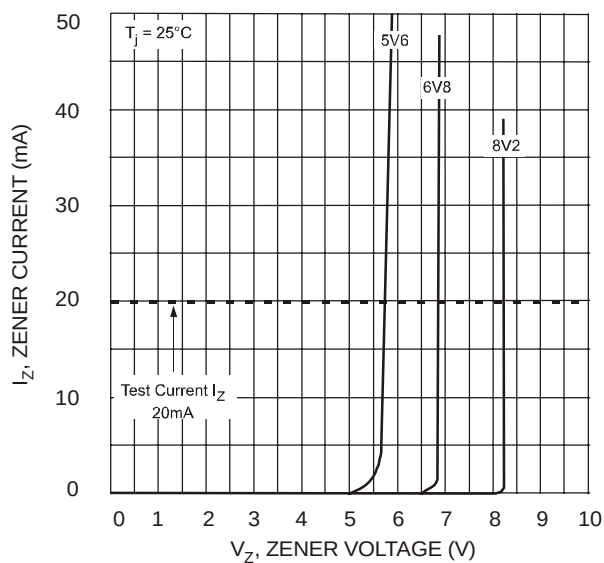


Fig. 4 Zener Breakdown Characteristics

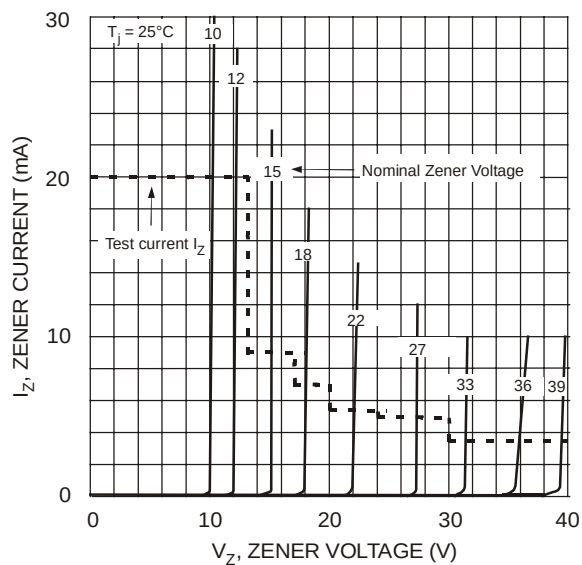


Fig. 5 Zener Breakdown Characteristics

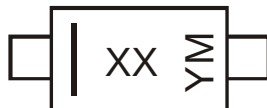
Ordering Information (Note 5)

Device	Packaging	Shipping
(Type Number)-7-F*	SOD-123	3000/Tape & Reel

*Add "-7-F" to the appropriate type number in Table 1 above example: 6.2V Zener = MMSZ5234B-7-F.

Notes: 5. For packaging details, go to our website at <http://www.diodes.com/datasheets/ap02007.pdf>.

Marking Information



XX = Product Type Marking Code (See Page 2)

YM = Date Code Marking

Y = Year (ex: N = 2002)

M = Month (ex: 9 = September)

Date Code Key

Year	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Code	J	K	L	M	N	P	R	S	T	U	V	W	X	Y	Z

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	1	2	3	4	5	6	7	8	9	O	N	D

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